

Merlin Yang

Oakland

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SUMMARY

I build digital experiences where design, performance, and engineering meet. I care obsessively about polish, responsiveness, and user delight. From fine-grained animations to clean, scalable architecture and AI-powered features that feel intuitive. I'm looking for a role where I can push the boundaries of full-stack engineering and ship products that feel as good as they look.

SKILLS

OpenAI API, Gemini API, Claude API, Prompt Engineering, LLM Evaluation, React 19, TypeScript, JavaScript, Tailwind, Vite, React Router, Node.js, Express, MongoDB, PostgreSQL, Supabase, Git, Vercel, Render

WORK EXPERIENCE

Fantast Bio Tech

Fullstack Dev Contractor

Jul 2024 - Aug 2025

Remote

- Developed AI-Assisted Social Media Post Generator full-stack SaaS application using React 19 + TypeScript, Express.js 5, and Supabase (PostgreSQL).
- Integrated OpenAI GPT API with sophisticated prompt engineering system (90+ instruction rules) for platform-aware content generation adhering to character limits (Twitter 280, LinkedIn 700, Instagram 2200).
- Built AI image generation pipeline using OpenAI's gpt-image-1 model with two-stage generation: AI-generated prompts from text content or custom user prompts.
- Engineered context-aware AI system incorporating company profile data (industry, values, products) with tone customization and AI-enforced style consistency.
- Implemented bilingual content generation system producing synchronized English and Chinese outputs with i18next internationalization.
- Deployed frontend on Vercel and backend on Render with JWT authentication, protected routes, and environment-based configuration.

PROJECTS

Metacognition Engine – AI-Powered Learning Platform

Full-Stack Developer

Sep 2025 - Dec 2025

- Developed full-stack web app using React 19, Express 5, and MongoDB with 20+ components, 15+ API endpoints, managing 1,200+ response documents.
- Architected multi-provider LLM abstraction layer using Factory Pattern, supporting 3 AI providers (Gemini, OpenAI, Claude) with runtime provider selection.
- Developed JSON parsing and validation layer for LLM outputs with fallback error handling across multiple model formats.
- Implemented three-tier question generation system creating goal-differentiated questions with expected key points
- Built confidence calibration analysis algorithm with threshold-based classification
- Built semantic evaluation engine using LLM APIs that scores responses (0-100), detects missing concepts, and generates constructive feedback.
- Implemented Google OAuth 2.0 authentication with Passport.js and session management via MongoDB store.

EDUCATION

Northeastern University

Master's in Computer Science

Sep 2025 - Jul 2027